

Basic Principles of Medicine 2

Nervous System & Behavioral Sciences

Directed Learning Activity

Learning Theory: Basics of Classical and Operant Conditioning



Photos created by Bing AI Image Generator

S2026

Developed by

Dr. Brenda Kirkby

bkirkby@sgu.edu

Department of Physiology, Neuroscience & Behavioral Sciences
St. George's University School of Medicine

Copyright

All year 1 course materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed, or published in any form, printed, electronic or otherwise. As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies. View only files, such as lecture recordings, are explicitly excluded from download and creating copies of these recordings by students and other users are strictly illegal. The author of this document has made the best effort to observe current copyright law and the copyright policy of St George's University. Users of this document identifying potential violations of these regulations are asked to bring their concern to the attention of the author.

The following objectives are covered in this DLA:

- SOM.MK.III.BPM2.4.NB.3.BEHS.0043 Explain the basic concepts and principles of classical conditioning, including its components, factors influencing its effectiveness, the extinction process, spontaneous recovery, and re-extinction
- SOM.MK.III.BPM2.4.NB.3.BEHS.0045 Explain the operant principles of reinforcement, punishment, extinction, extinction burst, and shaping
- SOM.MK.III.BPM2.4.NB.3.BEHS.0046 Identify the factors that influence the effectiveness of reinforcers and punishers
- SOM.MK.III.BPM2.4.NB.3.BEHS.0044 Describe the disadvantages of using punishment, including learned helplessness
- SOM.MK.III.BPM2.4.NB.3.BEHS.0047 Identify the different types of reinforcement schedules used to modify a behavior and explain the advantages and disadvantages of each type
- SOM.MK.III.BPM2.4.NB.3.BEHS.0048 Explain the role of classical and operant conditioning in psychoactive drug use and treatment, including the concepts of conditioned euphoria and conditioned tolerance

Recommended Readings

Current Diagnosis & Treatment: Psychiatry, 3e (2019), M.H. Ebert, J.F. Leckman, and I.L. Petrakis

- Chapter 10: Roots of Behavioral & Cognitive-Behavioral Interventions
- <https://accessmedicine-mhmedical-com.sgu.oclc.org/content.aspx?bookid=2509§ionid=200980307>

Table of Contents

Section 1: Behavioral Psychology Principles	4
Section 2: Classical Conditioning.....	4
General Principles	4
Factors Influencing the Effectiveness of Classical Conditioning	5
Classical Extinction	5
Section 3: Operant Conditioning.....	5
General Principles	5
Terminology	5
Factors Affecting the Effectiveness of Reinforcement and Punishment	6
Disadvantages of Punishment.....	6
Operant Extinction	6
Operant Schedules.....	7
Effect of Schedules on Speed of Learning and Extinction.....	7
Section 4: Practical Application of Behavioral Theory	7
Operant Conditioning in Substance Use	7
Classical Conditioning in Substance Use	8
Treatment of Substance Use Disorders using Behavioral Principles	8
Section 5: Concluding Statement.....	9
Section 6: Test Your Understanding	9
Questions	9
Answers and Explanations	11

Section 1: Behavioral Psychology Principles

The behavioral psychology model of learning assumes that most human behavior is learned through associations made while interacting with the environment. Two types of such associative learning are:

- **Classical Conditioning:** Association of a neutral stimulus with a stimulus that naturally and involuntarily produces a physiological response
- **Operant Conditioning:** Association of a behavior and its reinforcing or punishing consequence

Section 2: Classical Conditioning

General Principles

Learning through **classical conditioning** involves pairing of a **neutral stimulus** with a physiologically relevant stimulus so that the neutral stimulus is able to trigger the same response as the physiologically relevant stimulus. The terminology in this learning paradigm is as follows:

- **Unconditioned stimulus:** The physiologically relevant stimulus
- **Unconditioned response:** The response of the physiologically relevant stimulus
- **Neutral stimulus:** The stimulus that does not produce a physiologically relevant response
- **Conditioned stimulus:** The formerly neutral stimulus that now triggers a physiologically relevant response
- **Conditioned response:** The response of the conditioned stimulus

Example: Conditioned Taste Aversion

After becoming ill from eating potato salad made with spoiled mayonnaise, the person feels nauseous when confronted with potato salad. The elements of classical conditioning are:

Before Learning:

Unconditioned Stimulus (US)	→	Unconditioned Response (UR)
Spoiled mayonnaise		Nausea

Neutral Stimulus

(Potato salad)

Learning Phase: Pairing of **US** (spoiled mayonnaise) with **Neutral Stimulus** (potato salad)

After Learning:

Conditioned Stimulus (CS)	→	Conditioned Response (CR)
Potato salad		Nausea

Explanation: Potato salad was a neutral stimulus. However, because it was paired with a physiologically relevant stimulus (spoiled mayonnaise), potato salad masquerades as if it is physiologically significant and causes the same effect as if it were spoiled mayonnaise.

The CR may be elicited by stimuli that are similar to, but not exactly the same as, the original CS (e.g., becoming nauseous in response to any salad, not just potato salad). This is known as **stimulus generalization**. In contrast, the CR may only be elicited by the original CS (e.g., becoming nauseous in response to potato salad only). This is known as **stimulus discrimination**.

Factors Influencing the Effectiveness of Classical Conditioning

The effectiveness of classical conditioning relies on how **predictive** the CS is that something of biological significance will happen. The predictive value of the CS is enhanced by:

- Having the CS precede the presence of the US by a short time interval
- Increasing the number and consistency of pairings of the CS and US (i.e., more learning trials)

Classical Extinction

Classical extinction refers to *unlearning* the CR by eliminating the association between the US and CS. The extinction process involves repeated exposure to the CS in the absence of the US. In the conditioned food aversion example previously described, the person is repeatedly exposed to potato salad (CS) without spoiled mayonnaise (US). Over time, the CR (nausea) in response to the CS (potato salad) should diminish. Following extinction, the CR may spontaneously return in response to the CS (e.g., nausea may return when around potato salad). The re-emergence of the CR after extinction is called **spontaneous recovery**. The CR tends to **re-extinguish** quickly as long as the US and CS remain disassociated.

Section 3: Operant Conditioning

General Principles

Learning through **operant conditioning** involves the association between a behavior and its consequence. Based on the consequence, a behavior is either strengthened (reinforced) or weakened (punished).

Terminology

Reinforcement (RF): A behavior is reinforced when the consequence of the behavior makes it *more* likely that the behavior will be repeated. There are two types of RF:

- **Positive RF:** A **pleasurable** (desirable) stimulus is applied (e.g., praise, money, attention).
- **Negative RF:** An **aversive** (undesirable) stimulus is removed or avoided (e.g., exemption from chores; not having to take an examination).

Shaping (successive approximations) is the process of reinforcing actions in a gradual fashion to develop the desired behavior (e.g., a child who refuses to speak is reinforced if any audible sound is made, then only after a phoneme is verbalized, then only after a single word is spoken, and so forth, until the desired behavior of full speech is achieved).

Punishment (P): A behavior is punished when the consequence of the behavior makes it *less* likely that the behavior will be repeated. There are two types of P:

- **Positive P:** An **aversive** (undesirable) stimulus is applied (e.g., scolding spanking, nausea).
- **Negative P:** A **pleasurable** (desirable) stimulus is removed (e.g., privilege loss; removing attention/affection; being sent to “time out” – a room lacking pleasurable activities).

Note that the terms “positive” and “negative” do not refer to good or bad but whether something is given/applied (positive) or taken away/removed (negative) to change behavior.

Factors Affecting the Effectiveness of Reinforcement and Punishment

Learning through operant conditioning is enhanced when the **contingency** between a behavior and its consequence is clear. Contingencies are clearest when the consequence is **specified** and occurs **immediately** and **consistently** after a behavior. In behavioral medicine, the contingency between the consequences of some health risk behaviors (e.g., smoking) is diminished due to delayed consequences. Thus, such health consequences may not be effective punishers to weaken health risk behaviors.

Another factor that influences the effectiveness of a reinforcer or punisher is whether it is primary or secondary. Those that are *naturally reinforcing* (e.g., food, water, sex) or *naturally punishing* (e.g., pain, nausea) are **primary** and are subject to **satiation** (diminished effectiveness) with repeated use. For example, the primary reinforcer of “food” ceases to be rewarding if the person is no longer hungry. In contrast, **secondary** reinforcers and punishers acquire their reinforcing or punishing ability through *learning*. For example, one must learn that “money” or “an A grade” are desirable consequences (reinforcers) and that “losing money” and “an F grade” are undesirable consequences (punishers). Satiation is less likely with secondary reinforcers and punishers than with primary ones, but still occurs.

Disadvantages of Punishment

Use of punishment has several potential disadvantages:

- Someone who is positively punished through spanking may: 1) become fearful/resentful of the punisher; and/or 2) imitate the punishing behavior with another person (e.g., hit a peer).
- The selected punishment (e.g., scolding) may actually be reinforcing to the individual and thus inadvertently *increase* undesirable behavior. For example, a child who feels ignored may deliberately misbehave to gain parental attention.
- Individuals who are repeatedly punished may develop **learned helplessness** – a sense of powerlessness that results in passive behavior. For example, a person who persistently fails or gets reprimanded may stop trying, adopting the attitude, “Whatever I do is wrong or doesn’t work, so why try?” Individuals who develop learned helplessness may experience depressed mood or other forms of emotional distress.
- Punishment does not help strengthen desirable behaviors but is limited to weakening undesirable behaviors. **Reinforcement** is important to strengthen desirable behaviors.

Operant Extinction

Operant extinction refers to unlearning a behavior by withholding the reinforcer that used to follow the behavior. For example, a girl who used to get reinforced with laughter for telling jokes now stops telling jokes because the reinforcing laughter no longer occurs. Note that punishment also weakens behavior but using different methodology than extinction. In punishment, a behavior stops because a *different* consequence occurs (e.g., the girl stops telling jokes because she gets ridiculed or loses privileges).

When the operant extinction process is initiated, an **escalation** in the behavior may initially occur, known as the “**extinction burst**.” This occurs because the behavior that previously was effective in getting something desirable now no longer works, so the person tries harder. For example, the girl who is accustomed to laughter after telling jokes will tell more jokes if the laughter does not come. With persistence in withholding the reinforcer, the extinction burst is ultimately followed by extinction.

Operant Schedules

Operant schedules are the rules that determine how often a targeted behavior is reinforced (RF) or punished (P). A **continuous schedule** involves RF/P of *every* occurrence of the targeted behavior. Any schedule that is not continuous is an **intermittent schedule** and are categorized as follows:

- 1) **Ratio schedules (response-driven):** RF/P occurs after a certain number of **responses**:
 - a. **Fixed Ratio:** RF/P occurs after a **predictable** number of responses (e.g., a salesperson gets paid after every 3 magazine subscriptions sold).
 - b. **Variable Ratio:** RF/P occurs after an **unpredictable** number of responses (e.g., a gambler randomly wins money playing the slot machine on the 3rd, 8th, and 10th lever pull).
- 2) **Interval Schedules (calendar-driven):** RF/P occurs on a **specific day or time** (e.g., a salesperson gets paid on a certain date regardless of the number of magazine subscriptions sold).
 - a. **Fixed Interval:** RF/P occurs on a *predictable* date/time (e.g., monthly salary is paid on the 15th).
 - b. **Variable Interval:** RF/P occurs on an *unpredictable* date/time (e.g., monthly salary is paid on random days, such as on the 1st, 12th, 17th, or 21st).

Effect of Schedules on Speed of Learning and Extinction

The schedule used to reinforce a behavior is an important determinant of how quickly a behavior is learned and how difficult it is to extinguish.

- 1) **Learning:** A **continuous schedule** results in **faster** learning than an intermittent schedule. This occurs because the consistency of a continuous schedule makes the *contingency* between the behavior and the consequence clearer than an intermittent schedule.
- 2) **Extinction:** An **intermittent schedule** results in **slower (more resistance to) extinction** than a continuous schedule. This occurs because the inconsistency of getting a reward in an intermittent schedule keeps the person performing the behavior “just in case” the reward will be triggered by the next response (ratio) or next day/time (interval). Behaviors learned through **variable ratio** reinforcement (e.g., gambling) are **most resistant** to extinction.

Section 4: Practical Application of Behavioral Theory

Of high relevance to physicians, operant and classical conditioning play a role in psychoactive substance use. The following sections highlight some of the behavioral principles that promote drug use.

Operant Conditioning in Substance Use

Drug use can be **positively** and **negatively** reinforced:

- a) **Positive Reinforcement:** Most psychoactive drugs cause **euphoria**. This positive reinforcement (pleasurable mood state) increases the likelihood of further drug use.
- b) **Negative Reinforcement:** Unpleasant withdrawal symptoms may occur if drug use is stopped after prolonged use. If drug use resumes, these **unpleasant symptoms are reversed**, thus negatively reinforcing drug use.

Classical Conditioning in Substance Use

Through classical conditioning, drugs get associated with stimuli in the drug-using environment (e.g., sights, sounds, smells). Over time, these environmental stimuli trigger physiological changes associated with the drug. The elements of classical conditioning for this scenario are shown below:

Unconditioned Stimulus (US)	→	Unconditioned Response (UR)
Drug		Physiological changes
Neutral/Conditioned Stimulus (CS)	→	Conditioned Response (CR)
Environmental stimuli		Physiological changes (drug-like or drug-opposite effects)

The CR can **mimic** the drug's effect (e.g., euphoria) or be **opposite to** the drug's effect (e.g., dysphoria). Both drug-like CRs and drug-opposite CRs can promote further drug use.

- 1) **“Drug-like” CRs** are **similar to** the drug's effects but are typically less intense. For example, someone may feel mild euphoria when injecting saline into a vein because the injection procedure was associated with past heroin use. This phenomenon is known as **“conditioned euphoria.”** These attenuated drug-like CRs are unsatisfying and cause the person to crave more of the drug.
- 2) **“Drug-opposite” CRs** are **opposite to** the drug's effects. The CSs (environmental stimuli) become cues that signal imminent drug use and resultant physiological disruption. The CSs trigger drug-opposite (compensatory) physiological processes to protect against this homeostatic disruption. For example, cocaine (the US) causes **euphoria and increases pulse** (the UR). If a person sees a building where they used cocaine, the building (the CS) may trigger compensatory CRs of **dysphoria and decreased pulse** in anticipation of imminent drug use.
 - a) If the person does not use the drug, then compensatory CRs have “wrongly” induced homeostatic imbalance. Compensatory CRs are unpleasant (e.g., dysphoria and decreased pulse), which often prompts drug use to reverse the unpleasant state.
 - b) If the person does use the drug, then compensatory CRs are reversed by the drug's direct effects, and homeostasis is restored. However, this means that the person experiences minimal effects of the drug, known as **conditioned tolerance**. Implications of conditioned tolerance are:
 - i. In a **typical** drug-using environment (one *with* CSs to trigger compensatory CRs), the person will need a **larger drug dose** to achieve the desired level of intoxication.
 - ii. In a **novel** environment (one *without* CSs to trigger compensatory CRs), **failure of conditioned tolerance** may occur. Full effect of the larger dose is experienced because compensatory CRs did not minimize the drug's effect; risk of lethal overdose exists.

Treatment of Substance Use Disorders using Behavioral Principles

Treatment of substance use disorders relies extensively on **operant** and **classical conditioning** principles (i.e., behavioral therapy). For example, operant treatment approaches focus on rewarding healthy (drug abstinent) behavior and discouraging unhealthy (drug use) behavior. Classical treatment approaches focus on the extinction process (i.e., breaking the association between drug use and specific environmental stimuli) to help eliminate environmental triggers of drug use.

Section 5: Concluding Statement

The Learning Theory Flipped Classroom session associated with this DLA is devoted to analyzing learned behaviors in terms of their classical and operant conditioning components. The session will highlight the role of these principles in explaining salient medical phenomenon such as learned fears and avoidance behaviors and white coat hypertension.

Section 6: Test Your Understanding

(Answer key with explanations are on the last page)

Questions

1. A 6-year-old girl is brought to the physician by her mother because of a 4-week history of crying episodes. These episodes started after she accidentally wet her pants while playing with a group of friends on the playground. She felt embarrassed and started to cry. Now she feels embarrassed and cries whenever she sees the playground. Which of the following is most likely the unconditioned stimulus?
 - A. Her mother
 - B. Playground
 - C. Feeling embarrassed and crying
 - D. Group of friends
 - E. Wetting her pants
2. A 15-year-old girl is brought to the physician by her mother because of a 6-month history of a fear of horses. Her fear started after being thrown off a poorly trained horse. As part of her treatment, the physician recommends that she ride a well-trained horse every day for 30 minutes. Which of the following learning principles best describes the recommended approach by the physician to reduce her fear of horses?
 - A. Positive punishment
 - B. Negative punishment
 - C. Classical extinction
 - D. Operant extinction
 - E. Extinction burst
3. A 14-year-old girl is brought to the physician by her parents for a routine examination. She has been healthy over the past year but has developed the bad habit of keeping her room very untidy. The physician suggests that the parents reward her when she keeps her room tidy by taking away a task that she dislikes, such as washing the dishes in the evenings. This arrangement is implemented, and the girl now keeps her room tidy. Which of the following learning principles best explains the improvement in the girl's tidiness?
 - A. Positive reinforcement
 - B. Negative reinforcement
 - C. Positive punishment
 - D. Negative punishment
 - E. Operant extinction

4. A 6-year-old boy is brought to the physician by his father because of a 4-month history of increasingly disruptive behavior. He started having temper tantrums both at home and at school because he likes the attention he receives. As a result, the frequency of his tantrums has increased. The physician recommends that everyone stop giving him attention when he has a tantrum so that his tantruming behavior will diminish. Which of the following learning principles best explains the anticipated reduction in the boy's tantruming behavior?
- A. Positive reinforcement
 - B. Negative reinforcement
 - C. Positive punishment
 - D. Negative punishment
 - E. Operant extinction
5. A 30-year-old man comes to the physician for a follow-up examination. He has a 10-year history of nicotine addiction and has started a smoking cessation program. He is rewarded with a \$15 gift certificate if he goes a week without smoking. He is punished with a \$15 fine if he smokes during the week. Gift certificates and fines are given out by a staff member every Monday. Which of the following best describes the schedule of reinforcement and punishment that the treatment program is using?
- A. Continuous
 - B. Fixed ratio
 - C. Variable ratio
 - D. Fixed interval
 - E. Variable interval

Answers and Explanations

1) E

The unconditioned stimulus (US) is the biologically relevant (natural) stimulus that causes an unlearned (natural) response. In this case, “wetting her pants” would naturally cause a 6-year-old child to feel embarrassed and cry. “Wetting her pants” is the US, whereas “feeling embarrassed and crying” would be the unconditioned response (UR). The playground was the neutral stimulus. The playground (neutral stimulus) got associated with wetting her pants (US), so now seeing the playground causes her to feel the same way as wetting her pants. With this learned association, the playground is no longer a neutral stimulus but a conditioned stimulus (CS). The CS is now triggering the same response as the US. The UR and the conditioned response (CR) are both the same (“feeling embarrassed and crying”).

2) C

The girl learned to fear horses through classical conditioning. The neutral stimulus (horse) became a conditioned stimulus (CS) when the horse got associated with the unconditioned stimulus (US) of “being thrown off a horse.” Both the unconditioned response (UR) and the conditioned response (CR) are “fear.” The physician recommends reversing this learning process by breaking the association between the US (being thrown off a horse) and the CS (horses). This is done by exposing the girl to the CS (horses) without the US (being thrown off a horse) occurring. This is the process of classical extinction (i.e., reversing the learning that had taken place through classical conditioning).

3) B

The behavior in question is the girl’s tidiness. The consequence of that behavior is that something unpleasant (chores) will be removed. The goal is to strengthen (reinforce) tidiness. The parents are reinforcing the behavior of tidiness by removing something unpleasant. Thus, her tidiness is being *negatively* reinforced. If the parents “applied” something pleasant (e.g., money) to increase her tidiness, then the answer would be positive reinforcement. Punishment refers to weakening of a behavior. The question specifically is asking about increasing a behavior (tidiness).

4) E

The boy has been getting attention for his temper tantrums, and this attention has strengthened his behavior of tantruming. In other words, the boy’s temper tantrums have been developed because of positive reinforcement (operant conditioning). The physician recommends stopping the reinforcer that was used to develop the behavior (i.e., undoing the operant learning). This is the definition of *operant extinction*. Punishment would be correct if there was a consequence other than the former reinforcer ceasing. For example, if the boy was spanked after throwing a tantrum, then the answer would be positive punishment; if the boy had privileges removed after throwing a tantrum, then the answer would be negative punishment.

5) D

The reinforcer/punisher (gift certificate/fine) is released based on a calendar-entered date (i.e., Mondays). The patient’s behavior of smoking or not smoking will not shift the date that the reinforcer/punisher is dispensed. Thus, an interval schedule is being used. The date each week that the reinforcer/punisher is released is predictable (Mondays, not random days each week); thus, the schedule is “fixed” interval.